

Aashish Gupta

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SUMMARY

Software engineer with 2 years of experience building data-intensive backend systems and AI pipelines at Leena AI. Architected a batch ETL pipeline with **PySpark** and **Apache Iceberg** featuring **SCD Type 2** temporal modeling, partitioned writes, and source-to-sink reconciliation. Contributor to **Apache Iceberg**'s Python SDK. Designed an AI-powered matching system using **ElasticSearch HNSW** vector indexing with relevance scoring at scale. Engineered distributed job processing with **RabbitMQ** and **Redis**, reducing service memory by **25%** and infra cost by **15%**.

EXPERIENCE

Software Development Engineer I | Leena AI | Gurugram, India | July 2024 - Present

- **AI-Powered Matching (Big 4 Client):** Led design of a candidate-role matching system using **ElasticSearch** ANN indexing (**HNSW**), custom relevance scoring, and data quality validation across curated datasets; achieved **85%** match accuracy, reducing manual screening by **50%**.
- **LLM Evaluation Framework:** Built a regression-testing pipeline for production AI agents covering **tool-call accuracy**, response generation, and safety guardrails with LLM-as-judge scoring and flake detection, enabling pre-release certification of prompt changes.
- **MCP/A2A Server:** Architected Leena's Autonomous Agent server using FastAPI with streaming response delivery, connecting the orchestration layer to external platforms via **MCP** and **A2A** protocols with bot-scoped **OAuth 2.0**; enabled enterprise clients to use Leena agents within Microsoft 365 Copilot and SAP Joule from day one.
- **No-Code Form Builder:** Delivered a drag-and-drop form builder with smart defaults and validation rules, cutting setup errors by **60%** and reducing engineering time on form-related tickets by **~70%**.

Software Development Intern | Leena AI | Gurugram, India | January 2024 - June 2024

- Engineered backend services (Node.js, **MongoDB**, **RabbitMQ**) for the Flexi Benefits platform, handling enrollment, claims, and policy workflows; automated eligibility checks and reporting, reducing manual ops by **40%**.
- Built secure REST APIs for enterprise clients (Sony, Coca-Cola, HDFC) using **Node.js** and TypeScript, ensuring reliable third-party data synchronization.
- Offloaded scheduled jobs to **RabbitMQ** workers with **Redis** caching, eliminating CPU/memory spikes and cutting service memory by **~25%**; migrated modular jobs to **AWS Lambda**, reducing infra cost by **~15%**.
- Instrumented services with **Prometheus** and **Grafana** dashboards for API latency, error rates, and throughput monitoring; resolved production issues before client impact.

Co-Founder & Head of Engineering | Digisigns | Delhi, India | January 2023 - December 2023

- Co-founded a cross-platform digital signage platform (Android TV, Tizen); designed **FastAPI** + **WebSocket** services with fault-tolerant retries and local caching for media streaming, sync, and offline playback in low-connectivity environments.
- Tuned buffering, caching, and rendering logic for low-spec devices, improving playback smoothness by **~25%** and reducing visual glitches across deployments.

PROJECT

spark-lakehouse-etl: Batch ETL Pipeline | Python, PySpark, Apache Iceberg, Parquet, pytest | github.com/aashish-g03/spark-lakehouse-etl

- Built a batch ETL pipeline processing financial transactions with schema validation, bad-record quarantine, deduplication, and broadcast joins with a customer dimension, writing to partitioned **Apache Iceberg** tables.
- Implemented **SCD Type 2** merge logic using **Spark SQL** to track customer tier and address changes over time with null-safe change detection and surrogate key generation; added source-to-sink reconciliation comparing row counts and sums per day against a source manifest.

crewmate: Multi-Agent Orchestration CLI | TypeScript, Bun, POSIX IPC, MCP/ACP protocols | github.com/aashish-g03/crewmate

- Built an open-source CLI that orchestrates multiple AI coding agents (Claude, Gemini, Codex) in parallel with a lock-free filesystem task queue (**POSIX rename**), supervisor auto-restart, and orphan recovery.
- Implemented dual transport modes: spawn-per-task for isolation and persistent ACP (**JSON-RPC 2.0** over stdio) with session management, progress streaming, and cancel propagation.

EDUCATION

Delhi Technological University (Formerly DCE) | Bachelor of Technology in Computer Engineering | New Delhi, India | 2024 | 8.69/10

- Relevant Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Database Management Systems

OPEN SOURCE

Open Source Contributor | Apache Iceberg | May 2026 - Present

- Contributed type safety fix to pyiceberg expression constructors ([PR #3421](https://github.com/apache/iceberg/pull/3421)), resolving mypy incompatibility for downstream users.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, C++, Java

Data & Pipeline: Apache Spark, PySpark, Apache Iceberg, Parquet, Databricks, Snowflake, PostgreSQL, MongoDB, a RabbitMQ, SQL, AWS Lambda, Docker, CI/CD

AI & Search: ElasticSearch, Redis, Semantic Embeddings, Vector Search (HNSW/ANN), LLM Evaluation, MCP, A2A Protocol, OpenAI/Anthropic APIs

Observability & Tooling: Prometheus, Grafana, GitHub Actions, Prisma